

Full Stack Developer

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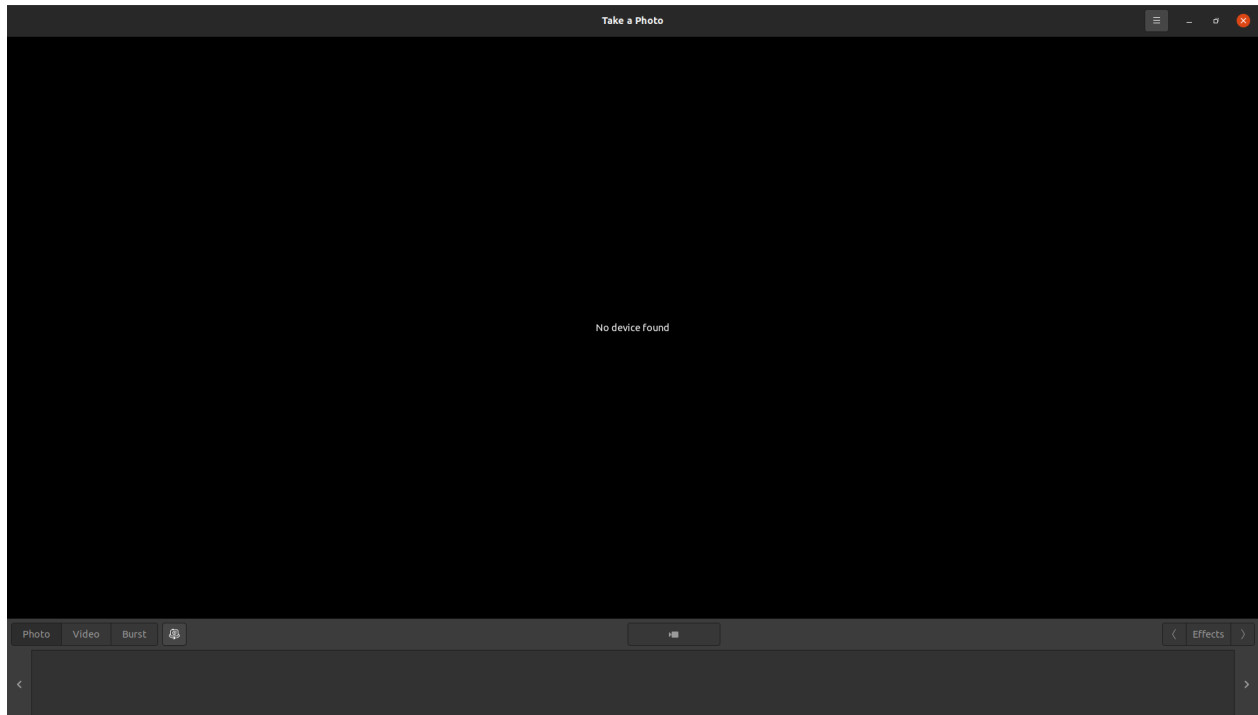
User Management System with Image Capturing App

Overview:

Goal is to build a User Management System using Angular for the frontend and Node.js for the backend. The application should include features for user authentication, role management, and an image capturing app.

Features:

1. **User Authentication:**
 - Supervisor and Worker will be able to login with the password provided by the Admin
 - Admin User will have default user name and password
2. **Admin Role:**
 - The system should have an "Admin" user by default.
 - Admin users should be able to create new accounts.
 - Admin users should be able to assign roles like "Supervisor" and "Worker" to other users.
 - Admin users should be able to remove user accounts.
3. **Role-based Access Control (RBAC):**
 - Control access to features based on the user's role.
 - Admin users should have access to all features.
 - Supervisor users should have limited access, and Worker users should have their own set of permissions.
4. **Image Capturing App:**
 - Implement an image capturing feature that accesses the inbuilt camera.
 - After user login, redirect them to a page where they can capture images using the inbuilt camera.
 - Store the captured images securely.



Milestones:

1. **Milestone 1: Authentication and User Management (Score: 30)**
 - Implement user registration and login functionalities.
 - Develop the admin user creation, role assignment, and account removal features.
2. **Milestone 2: Role-based Access Control (Score: 30)**
 - Implement role-based access control for different user roles.
 - Ensure proper handling of permissions for Admin, Supervisor, and Worker users.
3. **Milestone 3: Image Capturing App (Score: 30)**
 - Create a page for image capturing after user login.
 - Implement the image capturing feature using the inbuilt camera.
 - Store the captured images securely in the backend.
4. **Milestone 4: Documentation and Deployment (Score: 10)**
 - Provide clear instructions on how to run the application locally.

- Optionally, deploy the application on a cloud platform and provide the link.
- Include explanations for any challenges faced or specific decisions made during development.

Requirements:

- Use Angular for the frontend and Node.js for the backend.
- Implement a database to store user information, roles, and captured images.
- Ensure proper validation and error handling.
- Apply best practices for code organization, security, and performance.

Submission:

- Provide the source code for both frontend and backend.
- Include instructions on how to run the application locally.
- Optionally, deploy the application on a cloud platform of your choice and provide the link.

Evaluation Criteria:

Candidates will be evaluated based on the following criteria:

1. **Functionality (Score: 30):** Does the application meet the specified requirements?
2. **Code Quality (Score: 30):** Is the code well organized, readable, and maintainable?
3. **Security (Score: 20):** Are proper security measures implemented, especially for authentication and image storage?
4. **User Interface (Score: 10):** Is the UI intuitive and responsive?
5. **Documentation (Score: 10):** Is there clear documentation on how to run the application?

Important Notes:

- Feel free to use any additional libraries or frameworks if necessary.
- Use version control (e.g., Git) and make regular commits.
- If you face challenges or make certain decisions during development, provide explanations in the documentation.